SMART HOME NETWORK SECURITY DOCUMENT

Section 1 - Introduction

This document presents a comprehensive risk assessment of a typical **Smart Home IoT environment**. It identifies common network security risks affecting a simulated smart home setup consisting of various IoT devices, integrated sensors, and the internal home network.

The report outlines potential threats, vulnerabilities, impacts, and provides recommended security controls and mitigation strategies. It analyses risks from multiple angles, combining **network security, IoT security, cyber risk management, and governance** perspectives.

The purpose of this assessment is to evaluate the impact and likelihood of identified risks and deliver practical, actionable security controls. This report follows industry best practices and standards, including the **NIST Cybersecurity Framework**, **ISO 27001** principles, and **GDPR** data protection requirements, while maintaining a strong Governance, Risk, and Compliance (GRC) focus.

Section 2 – Smart Home Environment

Scenario:

A modern three-bedroom family home with the following connected devices and setup:

- Smart cameras (indoor/outdoor)
- Smart door lock
- Smart thermostat
- Smart lighting system
- Smart Tv and streaming devices
- Mobile apps and remote control
- Home Wi-Fi router (dual band)
- Cloud services for device management and data storage

- Smart Speaker / Voice Assistant (Amazon Alexa / Google Home)

All devices are connected and to a single home Wi-Fi network. Family members access the devices remotely using their smartphones. Some devices have default or weak passwords, and firmware updates are done manually.

This setup represents a typical real-world smart home used by many families today.

Section 3 – Identify Network and IoT Risks

The table below summaries the major risks identified in the smart home environment:

Risk	Threat	Impact	Likelihood	Risk Level	Recommended control
Weak/Default passwords	Unauthorized access to devices	Full device takeover, camera spying, lock manipulation	High	critical	Enforce strong unique passwords + MFA
Insecure device communication	Eavesdropping on unencrypted traffic	Data interception, privacy breach	High	High	Use WPA3 encryption and TLS 1.3
Excessive data collection	Privacy leak to cloud providers	Loss of personal data, profiling	High	High	Review permissions, disable unnecessary features
Lack of network segmentation	Lateral movement if one device is compromised	Compromise of entire home network	Medium	High	Create separate IoT VLAN/Guest Network
Outdated firmware	Exploitation of known vulnerabilities	Malware infection, botnet recruitment	Medium	High	Enable auto-update and regular patching

Phishing attacks on mobile apps	Credential theft	Account takeover and remote-control loss	Medium	Medium	Security awareness training
Exposed device ports and APIs	Remote exploitation	Unauthorized access	Medium	Medium	Disable UPnP and unused ports
No regular backups	Loss device configurations	Operational disruption	Medium	Medium	Regular configuration backups

Section 4 – Recommended Security Controls

To reduce identified risk to an acceptable level, the following security controls are recommended:

4.1 Access control

- Use strong unique passwords for every device and account
- Enable multifactor authentication (MFA) where available
- Apply the principle of least privilege
- Regularly review and revoke access for old devices

4.2 Network protection

- Implement network segmentation (separate IoT devices from main computers
- Use WPA3 encryption on the Wi-Fi routers
- Disable WPS and UPnP features
- Change default router admin credentials
- Consider using a firewall with IoT protection features

4.3 Devie & Firmware Management

- Enable automatic firmware updates
- Maintain and inventory of all IoT devices
- Replace or retire devices that no longer receive updates

4.4 Privacy and Data Protection

- Review and minimise data sharing with cloud providers
- Disable microphone and cameras when not in use
- Conduct periodic privacy impact assessments

4.5 user awareness and Monitoring

- Regular cybersecurity awareness for all residents
- Install reputable antivirus on computers and mobile devices
- Enable logging on the router and important devices
- Monitor connected devices regularly

Section 5 – conclusion

Smart home IoT devices offer great convenience but also significantly expand the attack surface of a home network. This assessment has shown that weak authentication, poor network segmentation, and privacy issues represent the most critical risks in a typical smart home environment.

By implementing the recommended security controls, particularly strong access management, network segmentation, and regular updates — the overall risk level can be substantially reduced. A proactive GRC approach helps homeowners enjoy the benefits of smart technology while maintaining adequate security and privacy.

This report demonstrates practical application of risk assessment methodologies that are highly relevant in both personal and professional cybersecurity environments.